

ERASTEEL TECHNICAL DATASHEET: ASP 2048

Application Family: Cold work | Standards: AISI M48 PM, W.-Nr. 1.3292

1. Grade Overview & Characteristics

ASP 2048 is a heavy-alloyed PM grade containing 9.8% Tungsten and 8.5% Cobalt. Provides extreme hardness retention and resistance against plastic deformation in heavy-duty cold work tooling and hard cutting.

2. Nominal Chemical Composition (wt. % range)

Element	Carbon (C)	Chromium (Cr)	Molybdenum (Mo)	Tungsten (W)	Vanadium (V)	Cobalt (Co)	Other
Range %	1.45 - 1.55	3.6 - 4.0	5.1 - 5.5	9.5 - 10.1	2.9 - 3.3	8.2 - 8.8	Super-high tungsten and cobalt

3. Mechanical Hardness & Property Profiles (Tempered state)

Metric	Value	Metric	Value (Rating 1-10)
Working Hardness Range	66.0 - 68.0 HRC	Machinability	4.5/10
Max Reachable Hardness	68.0 HRC	Wear Resistance	8.5/10
Annealed Delivery State	Max 280-350 HB	Toughness	5.5/10
Hot Hardness Retention	Secondary hardening	Hot Hardness	9.0/10
Heat Treatment Type	Vacuum / Salt bath	Grindability	5.0/10

4. Industrial Applications

Heavy cold stamping, fine blanking dies for abrasive alloys, thread rolling dies, cold extrusion punches, gear cutters.

5. Heat Treatment & Tempering Guidelines

Austenitizing at 1180-1210°C. Triple tempering at 560°C. Reachable working hardness 68 HRC.